Kim 2020 LUAD — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

One connected manifold (0 detached islands). Dominant sample P1028 (40%) spread throughout, not an archipelago; median 5 of 11 samples per 30-cell neighbourhood. iLISI 1.00→2.61 across the θ sweep (θ=10 on the mixing plateau). Read on the embedding + iLISI, not a scalar.

Gate 2 — module geography → GO (non-branching)

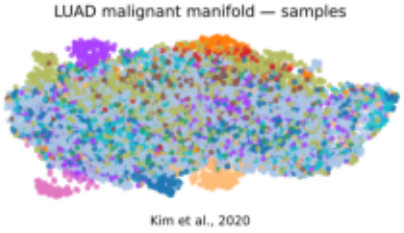
Non-branching: single developmental axis. Gast/pSom overlap (spatial r=−0.11); both oppose Crest (Gast-Crest r=−0.53, pSom-Crest r=−0.38). Crest is a terminal inverse gradient (Crest-high decile pseudotime 0.75; joint-high enrichment 0.31/0.44 < 1). Crest terminal to both modules — no Crest-arm inversion flag. PPT tree over-tiles the blob (no clean Y).

Gate 3 — hub cross-repression → NA

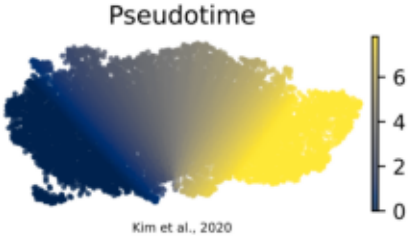
Not applicable — non-branching manifold has no Gast/pSom bifurcation hub to test.

STEP FIGURES

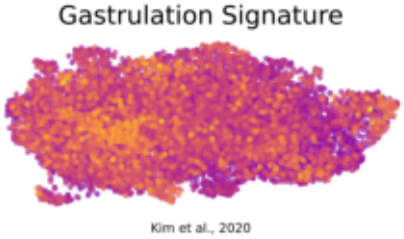
Manifold — samples



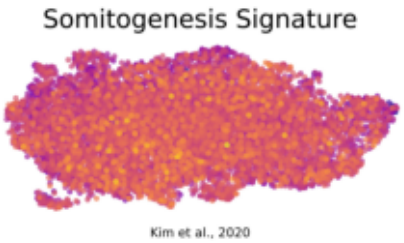
Pseudotime



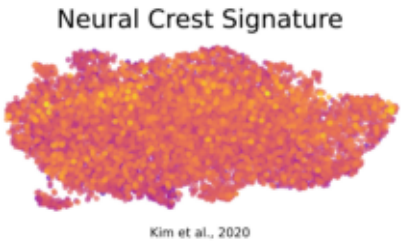
Gastrulation



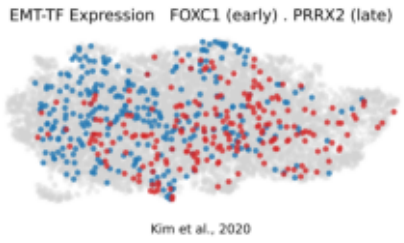
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Lung Adenocarcinoma (Kim et al., 2020), 10x — 11 patients; 10,538 malignant cells (10,368 after thresholding) enter the analysis; the whole malignant manifold enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA (LINEAR geometry). The embryonic-development programme reproduces as a single continuous gastrulation/somitogenesis → neural-crest axis: Gast/pSom co-vary at the origin, Crest rises to the terminus (inverse gradient), no competing-arm bifurcation — so Gate 3 has no hub to test. EMT-TFs FOXC1 (root) → PRRX2 (terminus).

Parameter choices

Min-cell threshold	≥100 cells/sample → 11/14 kept (no dominant cliff, top/2nd 1.2 → floor)	**Integration**	Harmony on sample, θ=10
Geometry	n_pcs=30; no scale before PCA	**Trajectory**	ElPiGraph curve N=20 (fold-free 2.89)
Root	Gast/pSom-high tip 0 (Crest-low origin)	**Module means**	Gast −0.040 · pSom +0.052 · Crest −0.045
EMT-TF axis	FOXC1 (root) → PRRX2 (terminus)		